Question 1

Create an Nginx service with 3 replicas, expose it on port 80, verify access from a browser, then scale it to 5 replicas.

Step 1: Initialize Swarm

Explanation: Creates a Docker Swarm cluster and makes the current machine the Manager Node.

docker swarm init

```
PS C:\WINDOWS\system32> docker swarm init
Swarm initialized: current node (msgtivvpl02ng3nszofbxtub6) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-2a1tjcs3mgkbzx8ucfqqrmskoe5x0ys8vqkql76v559vaj9yi4-6rsdb2y3o331x
9n6pkduyfwyv 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

Step 2: Create Service

Explanation: Creates an Nginx service with 3 replicas and publishes port 80.

docker service create --name nginx-service --replicas 3 -p 80:80 nginx

```
PS C:\WINDOWS\system32> docker service create --name nginx-service --replicas 3 -p 80:80 nginx
jhckmx9pu4i9su2kwuo21duar
overall progress: 3 out of 3 tasks
1/3: running
2/3: running
3/3: running
verify: Service jhckmx9pu4i9su2kwuo21duar converged
```

Step 3: Verify Service

Shows running services and replica count.

docker service ls

```
PS C:\WINDOWS\system32> docker service ls
ID             NAME            MODE         REPLICAS   IMAGE          PORTS
jhckmx9pu4i9   nginx-service   replicated   3/3        nginx:latest   *:80->80/tcp
PS C:\WINDOWS\system32> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS
                NAMES
ff43873bb2ae   nginx:latest   "/docker-entrypoint.…"   38 seconds ago   Up 37 seconds   80/tcp
                nginx-service.2.kf2f4tv0nrcko2oeqkgrip1te
7b23bb65fae7   nginx:latest   "/docker-entrypoint.…"   38 seconds ago   Up 37 seconds   80/tcp
                nginx-service.1.ct62tx0jgm48fiwmxyuaq531q
a630e0ff88eb   nginx:latest   "/docker-entrypoint.…"   38 seconds ago   Up 37 seconds   80/tcp
                nginx-service.3.j6tiii19dxhhnud10m9ofy10j
0cac1eed2d42   server         "make run"               2 years ago      Up 6 seconds    0.0.0.0:80->80/
tcp, 0.0.0.0:60886->5432/tcp   server-instances
```

Step 4: Scale Service

Increases replicas from 3 to 5.

docker service scale nginx-service=5

```
PS C:\WINDOWS\system32> docker service scale nginx-service=5
nginx-service scaled to 5
overall progress: 5 out of 5 tasks
1/5: running
2/5: running
3/5: running
4/5: running
5/5: running
verify: Service nginx-service converged
```

Step 5: Verify Scaling

Displays all running tasks.

docker service ps nginx-service

```
PS C:\WINDOWS\system32> docker service ps nginx-service
ID             NAME              IMAGE          NODE             DESIRED STATE   CURRENT STATE                ERROR     PORTS
ct62tx0jgm48   nginx-service.1   nginx:latest   docker-desktop   Running         Running 8 minutes ago
kf2f4tv0nrck   nginx-service.2   nginx:latest   docker-desktop   Running         Running 8 minutes ago
j6tiii19dxhh   nginx-service.3   nginx:latest   docker-desktop   Running         Running 8 minutes ago
9gm2wirggc85   nginx-service.4   nginx:latest   docker-desktop   Running         Running about a minute ago
l3lhn70lzuc5   nginx-service.5   nginx:latest   docker-desktop   Running         Running about a minute ago
```

Cleanup

Remove Service

docker service rm nginx-service

```
PS C:\WINDOWS\system32> docker service rm nginx-service
nginx-service
PS C:\WINDOWS\system32> docker service ps nginx-service
no such service: nginx-service
```

Leave Swarm

docker swarm leave --force

```
PS C:\WINDOWS\system32> docker swarm leave --force
Node left the swarm.
```

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Question 2

Que: Create a service with 2 replicas and inspect where the containers are running.

Step 1 docker swarm init

```
PS C:\WINDOWS\system32> docker swarm init
Swarm initialized: current node (lxrwldnoy0bysfs53r7bzyrh5) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-068gl71vyr05ndrb80xw75jf1uqypzae7e4m6esuls3ngdv7jo-bjmcdhw

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions
```

Step 2 docker service create --name web --replicas 2 nginx

Creates a service with 2 containers.

```
PS C:\WINDOWS\system32> docker service create --name web --replicas 2 nginx
y7ou1baa49334619phb3wobi6
overall progress: 2 out of 2 tasks
1/2: running   [==================================================>]
2/2: running   [==================================================>]
verify: Service y7ou1baa49334619phb3wobi6 converged
```

Step 3:

Displays tasks and the node on which each container is running.

docker service ps web

```
PS C:\WINDOWS\system32> docker service ps web
ID             NAME      IMAGE          NODE             DESIRED STATE   CURRENT STATE            ERROR     PORTS
n3nyt8bwo5dl   web.1     nginx:latest   docker-desktop   Running         Running 45 minutes ago
m6sv451r9gtb   web.2     nginx:latest   docker-desktop   Running         Running 45 minutes ago
```

Cleanup

Remove Service: docker service rm web

Leave Swarm: docker swarm leave --force

```
PS C:\WINDOWS\system32> docker service rm web
web
PS C:\WINDOWS\system32> docker swarm leave --force
Node left the swarm.
```

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Question 3

Que: Simulate node maintenance by draining a worker node.

Step 1: docker node ls

Lists all nodes.

```
PS C:\Users\PG-ITISS> docker node ls
ID                            HOSTNAME         STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VERSION
a4720uxhiteacv5a1gsodi5th *   docker-desktop   Ready     Active         Leader           29.4.0
```

Step 2: Marks worker1 unavailable for scheduling.

docker node update --availability drain <worker1>

```
PS C:\Users\PG-ITISS> docker node update --availability drain docker-desktop
docker-desktop
```

Step 3: Observe tasks moving to another node.

docker service ps nginx-service

Step 4 (Restore Node): Makes the node available again.

docker node update --availability active worker1

Cleanup

Remove Service: docker service rm nginx-service

Leave Swarm: docker swarm leave --force

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Question 4

Que: Perform a rolling update from Nginx latest to Nginx alpine.

Step 1 docker swarm init

```
PS C:\Users\PG-ITISS> docker swarm init
Swarm initialized: current node (ahibt04lhm4u6kv8ne5ewkotj) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-2jtp3pjgjxi67q2kz8nmunou9loew1y1ogd2uxe414zpcuebbk-c4yw2d8r2gvyndlz
wjqxom9e2 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

Step 2 docker service create --name web --replicas 3 nginx

```
To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
PS C:\Users\PG-ITISS> docker service create --name web --replicas 3 nginx
tazgxks5vo25eld55sbr08d83
overall progress: 3 out of 3 tasks
1/3: running
2/3: running
3/3: running
verify: Service tazgxks5vo25eld55sbr08d83 converged
```

Step 3 docker service update --image nginx:alpine web

Updates containers one by one.

```
PS C:\Users\PG-ITISS> docker service update --image mginx:alpine web
web
overall progress: 0 out of 3 tasks
1/3: preparing
2/3:
3/3:
service update paused: update paused due to failure or early termination of task 1i3qyvictcal2iymzq2zoy818
```

Step 4 docker service ps web

Verify updated containers.

```
PS C:\Users\PG-ITISS> docker service ps web
ID             NAME         IMAGE          NODE             DESIRED STATE   CURRENT STATE                      ERROR
                            PORTS
55r1q5b80g1w   web.1        nginx:latest   docker-desktop   Running         Running 2 minutes ago
wfwq4t4cmfsy   web.2        mginx:alpine   docker-desktop   Ready           Preparing less than a second ago
bd5xr01qjx5o    \_ web.2    mginx:alpine   docker-desktop   Shutdown        Rejected 4 seconds ago             "No s
uch image: mginx:alpine"
sep62vuvs2fx    \_ web.2    mginx:alpine   docker-desktop   Shutdown        Rejected 9 seconds ago             "No s
uch image: mginx:alpine"
ek4rp6erdr9r    \_ web.2    mginx:alpine   docker-desktop   Shutdown        Rejected 14 seconds ago            "No s
uch image: mginx:alpine"
618opl1aa5pi    \_ web.2    mginx:alpine   docker-desktop   Shutdown        Rejected 19 seconds ago            "No s
uch image: mginx:alpine"
i51pf04dvisw   web.3        nginx:latest   docker-desktop   Running         Running 2 minutes ago
PS C:\Users\PG-ITISS>
```

Concept Learned:

Rolling updates provide near-zero downtime deployment.

Cleanup

Remove Service: docker service rm web
Leave Swarm: docker swarm leave --force

```
i51pf04dvisw   web.3> docker service rm webocker-des
webC:\Users\PG-ITISS>
PS C:\Users\PG-ITISS> docker swarm leave --force
Node left the swarm.
```

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Question 5

Que: Create an overlay network and attach a service to it.

Step 1 docker swarm init

```
PS C:\Users\PG-ITISS> docker swarm init
Swarm initialized: current node (ayjdx7yyew9xhf75tmjl22gms) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-52b45b56tgovaqs0ejbbnud7zma58s56vprpx19ex5h9gszuaw-8uv3go1y54833kwefq6lljo43
 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

Step 2 Creates a swarm-wide overlay network.
docker network create -d overlay mynet

```
PS C:\Users\PG-ITISS> docker network create -d overlay mynet
l8j9gbva1iwtfjt8v8h9ga0nj
```

Step 3 Deploys the service on the overlay network.
docker service create --name web --network mynet nginx

```
PS C:\Users\PG-ITISS> docker service create --name web --network mynet nginx
oa0dek57murnpgzp0vswgj56j
overall progress: 1 out of 1 tasks
1/1: running   [==================================================>]
verify: Service oa0dek57murnpgzp0vswgj56j converged
```

<u>Step 4</u> Displays network information.

docker network inspect mynet

```
PS C:\Users\PG-ITISS> docker network inspect mynet
[
    {
        "Name": "mynet",
        "Id": "l8j9gbva1iwtfjt8v8h9ga0nj",
        "Created": "2026-06-06T07:33:19.669567098Z",
        "Scope": "swarm",
        "Driver": "overlay",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "10.0.1.0/24",
                    "Gateway": "10.0.1.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.driver.overlay.vxlanid_list": "4097"
        },
        "Labels": {},
        "Peers": [
            {
                "Name": "f080ad2f73d0",
                "IP": "192.168.65.3"
```

Concept Learned: Overlay networks allow communication between containers on different swarm nodes.

Cleanup

Remove Service: docker service rm web
Remove Network: docker network rm mynet
Leave Swarm: docker swarm leave --force

```
PS C:\Users\PG-ITISS> docker service rm web
web
PS C:\Users\PG-ITISS> docker network rm mynet
mynet
PS C:\Users\PG-ITISS> docker swarm leave --force
Node left the swarm.
```

Quick Memory Trick

Action	**Command**
Initialize Swarm	**docker swarm init**
Create Service	**docker service create ...**
List Services	**docker service ls**
Inspect Tasks	**docker service ps <service>**
Scale Service	**docker service scale <service>=5**
Update Service	**docker service update ...**
Remove Service	**docker service rm <service>**
Leave Swarm	**docker swarm leave --force**

This sequence—Init → Create → Verify → Modify → Verify → Remove → Leave—works for most Docker Swarm lab exams.